

A short-term power prediction method based on numerical weather prediction correction and the fusion of adaptive spatiotemporal graph feature information for wind farm cluster

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ABSTRACT

In recent years, with the construction of new electricity system, the scale of wind power grid connection has been continuously expanding. Accurate wind power prediction is crucial for ensuring the stable operation of the power system. However, the existing short-term power prediction methods for wind farm cluster (WFC) have shortcomings in extracting spatiotemporal features, making it difficult to achieve high-precision prediction. Therefore, this paper proposes a WFC short-term power prediction method based on numerical weather prediction (NWP) correction and the information fusion of adaptive spatiotemporal graph feature for WFC. Firstly, similar information between future and past periods of NWP is sought to achieve NWP wind speed correction by the defined error correlation ratio. Then, the corrected NWP wind speed is combined with the information of geographical location of the wind farms, and the adaptive spatiotemporal graph network is used to achieve short-term wind power prediction of the WFC. The proposed method is applied to two large-scale WFCs in China. The experimental results show that compared with other comparative algorithms, the RMSE is reduced by 0.0100 averagely and the MAE is reduced by 0.0096 averagely. The coefficient of determination is improved 0.0440 averagely and the accuracy rate is improved 1.01% averagely. The above results indicate that the predicted power obtained by the proposed method is closest to the actual power. It is beneficial for improving the utilization efficiency of wind power and reducing the pressure on power system scheduling.

1. Introduction

1.1. Background

In recent years, with the increasing depletion of fossil fuels, more and more countries have regarded renewable energy as the mainstay of power generation (Wang et al., 2023; Zhang et al., 2022; Zhuo et al., 2022). Wind energy is widely used due to its green and renewable characteristics, and the installed capacity of wind power continues to increase (Huang et al., 2024; Huang et al., 2023). Fig. 1 shows the changes in installed capacity of wind power from 2014 to 2023. In addition, wind power generation is mainly affected by weather and environmental factors, and it has the characteristics of randomness, intermittency, and volatility (Yang et al., 2025; Lu et al., 2024). If large-

scale wind power is directly integrated into the power grid, it will affect the stability of the power system and increase scheduling difficulties (Yang et al., 2025; Lu et al., 2024). Accurate wind power prediction can predict the trend and fluctuation of wind power in advance, which is of great significance for the formulation of dispatch plans of power grid and the stable operation of the power system (Yang et al., 2024; Lu et al., 2023).

1.2. Previous work

Wind power prediction (WPP) establishes a prediction model based on meteorological and environmental information of wind farms (WFs) (Yang et al., 2024; Li et al., 2024). The existing WPP methods can be roughly divided into three categories: physical models, statistical

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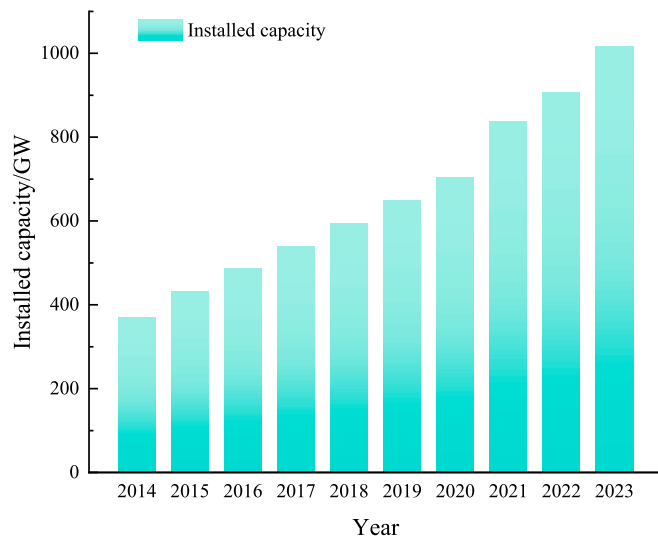


Fig. 1. Wind power installed capacity change chart from 2014 to 2023.

models, and artificial intelligence models. Physical method mainly uses physical laws that control atmospheric behavior to simulate the airflow field of wind farms, and then calculate the corresponding power generation of wind turbines. Numerical weather prediction (NWP) plays a crucial role in physical methods. It is widely used for wind power prediction on medium and long-term time scales. However, the modeling process of Physical method is complex and its calculation speed is slow (Yang et al., 2025; Li et al., 2024). So, it is rarely used for short-term wind power prediction. Statistical methods establish predictive mapping relationships based on historical data (Yang et al., 2024; Yang et al., 2024). However, statistical methods are not suitable for solving power prediction problems with nonlinear data. In the past decade, artificial intelligence models have flourished in various fields such as time series prediction, fault diagnosis, quality inspection, and structural safety assessment (Wang et al., 2023). Artificial intelligence models can capture the nonlinear relationship between the original data samples and WP, fully explore the data features in the original data samples, and improve the prediction accuracy of WP (Moreno, et al., 2024). However, a single statistical machine learning model has drawbacks such as local optima, overfitting, and poor generalization ability. When predicting the regions with the most drastic sample fluctuations, the prediction error is relatively large (Ribeiro, et al., 2024). Based on the above issues, combination prediction models are gradually being applied in the field of WPP. The combination prediction model is based on maximizing information utilization (Takara et al., 2024). While considering the advantages of different models comprehensively, it optimizes the shortcomings of each model, significantly improving the accuracy of WPP (Li et al., 2024).

For short-term wind power prediction, NWP wind speed is a very important data feature, and the accuracy of NWP wind speed will directly affect the short-term wind power prediction results (Ahmadi and Khashei, 2021). Therefore, experts and scholars have done a lot of work in NWP wind speed correction and wind speed prediction. Samadianfard et al (2020) used Whale Optimization Algorithm to optimize multi-layer perceptron (MLP-WOA) for wind speed prediction. Although this method achieves wind speed prediction, it requires careful parameter design to avoid overfitting of the model. Sinvaldo Rodrigues Moreno et al. (2020) & Silva et al. (2021) used decomposition methods for time series such as Variational Mode Decomposition (VMD) and Singular Spectrum Analysis (SSA) to decompose the wind speed series, further mining the data features of wind speed. Then, the accuracy of wind speed prediction is improved by integrating multiple deep learning algorithms into an artificial intelligence model for wind speed prediction. Duan et al. (2021) decomposes the prediction error to correct the

previously predicted wind speed, improving the accuracy of wind speed prediction. However, prediction errors have significant randomness, which may lead to unstable correction performance. Jiang et al. (2021) proposes a combined prediction model of deterministic prediction based on improved multi-objective optimization algorithm and interval prediction based on distribution fitting, which can provide effective wind speed point prediction and interval prediction. Takara et al. (2024) proposes a combined prediction model that combines advanced deep neural networks with superposition-based probability learning. This method can improve the multi-step prediction accuracy of wind speed, laying the foundation for effective energy management and policy planning in the renewable energy industry. Aly (2020) applied an intelligent hybrid model of neural wavelets, time series, and recursive Kalman filter to wind speed prediction to improve the accuracy of wind speed prediction. However, the recursive Kalman filter has disadvantages such as sensitivity to initial values and strong dependence on noise statistical characteristics. In addition, artificial neural networks have the disadvantages of long training time and poor interpretability.

In terms of wind power prediction for wind farm clusters, Wu et al (2021) proposed a CNN-LSTM hybrid model and the spatiotemporal correlation vectors of meteorological features are extracted using Convolutional neural network (CNN) and the time series are reconstructed. For multi-step prediction, Long Short Term Memory (LSTM) uses reconstructed time series as input to extract temporal feature relationships between historical time points. The test results show that the hybrid model outperforms a single CNN and a single LSTM model in prediction. Yildiz et al. (2021) proposed an improved deep convolutional neural network based on residuals for efficient WPP. This network has lower complexity and computational cost, but the remaining connection methods also lose some features of the data. Heinemann et al. proposed a combined prediction model of equal weight decision tree and Support vector machine (SVM), which improved the prediction accuracy of WP by 37 % (Heinemann and Kramer, 2016). Hossain et al. constructed three models of LSTM, Gate recurrent unit (GRU), and fully connected networks as well as the Harris Hawks algorithm is used to determine the combined weights of the three models, improving modeling efficiency (Hossain, et al, 2021). Alkessaiberi et al. (2022) used Bayesian algorithms to optimize the parameters of the Deep learning (DL) model to improve modeling accuracy, and the lag components are introduced into the input of the data to improve the model's learning ability for dynamic data. Samadianfard et al. (2020) provides a prediction method based on enhanced residual CNN, which reduces computational costs by converting wind speed sequences and converting meteorological data into images. Altan et al. (2021) discussed the accurate prediction of wind speed for non-stationary effects using the Grey Wolf Optimizer based on Long Short Term Memory (LSTM-GWO). The smoothed data is filtered to determine any missing data during the measurement process. The results indicate that the model improves the original model and it better reflects the nonlinear characteristics of wind speed. However, it does not provide equal support for traditional resources.

Several deep learning based spatial methods have been proposed to extract spatial correlations (Zhao et al., 2024; Wen et al., 2022; Wang et al., 2023; Xu et al., 2024). Wang et al. (2022) proposed an ultra short-term power prediction method for WFCs based on a constructed hierarchical directed graph structure. This method can describe the spatio-temporal correlation of adjacent WFs and exhibits excellent prediction accuracy. Pei et al. (2022) proposed a prediction model for regional WFs that can dynamically analyze the spatial correlation of time delay between regional WFs and other WFs based on the physical characteristics of wind speed time delay within the region. Yu et al. (2020) proposed a Stacked Graph Neural Network (SGNN) for extracting spatial features between different wind turbines within a certain range of wind farms based on geographic location and other relevant graphic information to improve prediction accuracy. Fu et al. (2019) proposed a multi-lead self-attention method to capture spatial correlations between WFs and it can

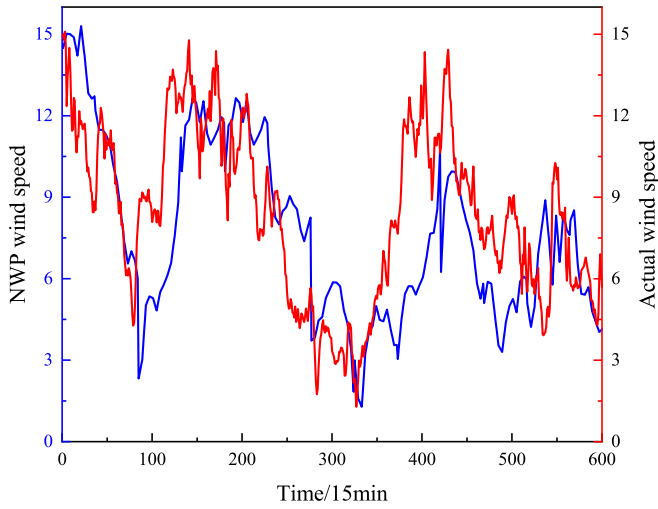


Fig. 2. Comparison between NWP wind speed and actual wind speed.

extract temporal dependencies by sequence to sequence. However, these methods were aimed at exploring the spatial correlation between different wind farms or wind turbines, without considering the spatial characteristics between WP variables for each wind turbine, which also have a significant impact on the output of WP. In recent years, graph neural networks have gradually begun to be applied in spatiotemporal prediction due to their ability to better handle unstructured data and effectively extract spatial features (Bentsen et al. 2023). Khodayar and Wang (2019) extracted spatial and temporal features by graph convolutional networks and LSTM respectively. Every WF is represented by nodes in an undirected graph. In addition, Bidirectional Gate Recurrent Unit (BiGRU) has stronger learning ability than LSTM because it can learn time series from both positive and negative directions simultaneously. Thereby, more important information time for related time series can be obtained (Wang et al., 2022). Wang et al. (2022) described the spatiotemporal correlation between wind farms by a hierarchical directed graph structure. Then the Graph Convolutional Network (GCN) is applied to predict the power of WFC. Zhao et al. (2024) propose a Double Attention-based Spatial-Temporal Neural Network (DA-STNet) for medium-term wind power forecasting. The attention mechanism is combined with graph neural network to improve the prediction accuracy of wind power. Peng et al. (2024) & Liu et al. (2023) applied graph neural networks to wind power ramping prediction and urban wind speed prediction, improving prediction accuracy and providing a new way to ensure power supply.

In summary, the decomposition methods of time series can further explore the hidden features of wind speed data. At the same time, the data of wind speed can identify the trend changes of wind speed series after decomposition. Some artificial intelligence algorithms also have long training times and cannot accurately predict the detailed fluctuations of wind speed. In terms of wind power prediction, graph neural networks can mine spatial correlation information between wind farms and it can be integrated with attention mechanisms and other feature engineering algorithms, demonstrating good portability. However, wind power may have different spatiotemporal correlation characteristics in different time scales, and NWP information and wind power information in multiple time scales can provide more data features for prediction models. Static graph neural networks are unable to capture dynamic information of wind power in multiple time scales, and dynamic graph neural network modeling suffers from problems such as long running time and low operating efficiency.

1.3. Our contributions in this article

In response to the above issues, the contributions made in this article

are as follows:

- (1) A new correction method of NWP wind speed is proposed. A new evaluation index that is named the error correlation ratio (ECR) is defined. The method is to find the minimum of the ECR is used to find similar segments between the original NWP wind speed and the historical actual wind speed (WS). And historical actual WS is instead of future predicted wind speed in similar segments to improve the accuracy of NWP and further enhance the accuracy of WPP.
- (2) A short-term wind power prediction model is proposed based on adaptive spatiotemporal graph network for wind farm cluster, which can integrate the information of NWP feature information and the information of wind power feature in multiple time scales, fully explore the spatiotemporal correlation between data features and wind power to achieve short-term wind power prediction for wind farm cluster.
- (3) The simulation experiment verifies the progressiveness and superiority of the proposed method by two WFCs from different regions of China.

The rest of this paper is organized as follows: in Section 2, the NWP wind speed correction and a short-term power prediction method based on the fusion of adaptive spatiotemporal graph feature information for wind farm cluster are proposed. In Section 3, several comparative experiments are conducted to analyze the experimental results and verify the effectiveness of the model proposed in the article by applying different evaluation metrics. Finally, conclusions are provided in Section 4.

2. Method

2.1. The description of the structure topology diagram and dataset

The graph network composed of the entire WFC is defined by G , where V represents the set of WFs, and N represents the number of vertices in the topology map, which can also be referred to as nodes in the topology map. E represents the set of edges with spatial proximity between nodes. A is the adjacency matrix of G , where $A \in \mathbb{R}^{N \times N}$. A_{v_i, v_j} indicates the connection between nodes v_i and v_j . The sequence of wind power can be represented as $X_t = (x_t^1, x_t^2, \dots, x_t^N)^T \in \mathbb{R}^{N \times F}$, where F represents the feature dimension, t represents the time step, and x_t^i represents the observed value of the i th wind farm at time step t .

A topology matrix $X = (X_{t-\tau+1}, \dots, X_{t-1}, X_t) \in \mathbb{R}^{\tau \times N \times F}$ with a given historical time step is given. And the future WP $\hat{Y} = (\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+M}) \in \mathbb{R}^{M \times N \times F}$ for M time steps is predicted. Therefore, the problem of WPP is further quantified as learning the mapping function F to map the topology information matrix that is for historical time step to the topology information matrix that is for future time step:

$$(\hat{X}_{t+1}, \hat{X}_{t+2}, \dots, \hat{X}_{t+M}) = F_{\theta}(X_t, X_{t-1}, \dots, X_{t-\tau+1}; G) \quad (1)$$

where, θ represents all learnable parameters in the model.

2.2. NWP wind speed correction

2.2.1. NWP wind speed characteristics

The information of the NWP is necessary for predicting WP. However, it often has significant deviations from actual weather information. The most important feature of NWP is the predicted wind speed. As shown in Fig. 2, there is a significant deviation between NWP wind speed and actual wind speed in known historical data. Firstly, NWP wind speed cannot track the trend of actual wind speed in a timely manner. Secondly, the forecast curve of NWP wind speed is relatively smooth and lacks detailed fluctuation information. To achieve the correction of NWP

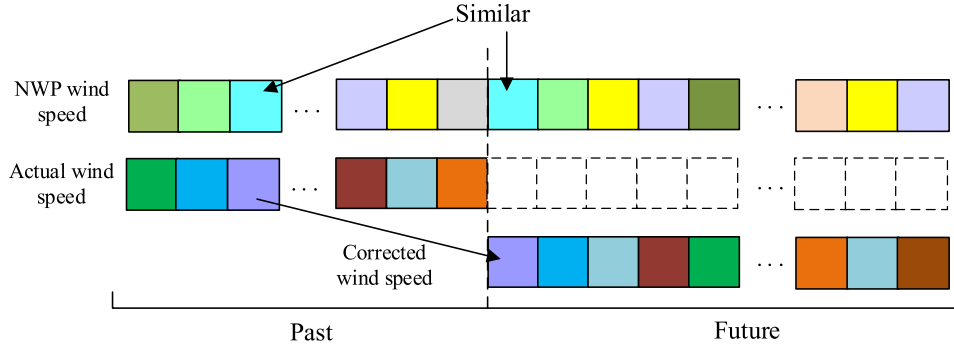


Fig. 3. NWP wind speed correction principle.

wind speed in the future period, a method of searching for similar segments was adopted for wind speed correction. Fig. 3 shows the principle for NWP wind speed correction.

The data of NWP wind speed has strong randomness and volatility, and its sequence contains not only trend information but also a large amount of random noise. If similar segments are directly searched for through the original sequence of NWP wind speed, it may cause misjudgment because the trend information of NWP wind speed is inability to be accurately determined. It results in the inability to correct NWP wind speed or poor effectiveness of NWP wind speed. Therefore, it is crucial to obtain the trend sequence of the wind speed from the original NWP wind speed. Firstly, the original sequence of wind speed is decomposed. Then the decomposed permutation entropies are calculated. Finally, the decomposed data is reconstructed based on PE to obtain the reconstructed trend component.

2.2.2. The decomposition of NWP wind speed

To improve the accuracy of signal decomposition and prevent the influence of noise on signal components, improved ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) signal decomposition method is used to decompose the original sequence of NWP wind speed. The ICEEMDAN method is a further improvement on the complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) method. ICEEMDAN solves the possible problems of pseudo modes in the CEEMDAN method by selecting white noise. This method not only solves the problem of mode aliasing but also greatly reduces residual noise in the intrinsic mode function (IMF). Accurate modal decomposition results can be obtained with fewer iterations, and they have been applied in various fields. It also has good decomposition and feature extraction capabilities for the information of different scale components contained in the time series of WP. The basic steps for decomposition using ICEEMDAN are as follows:

Step 1: Set x as the time series of WP that needs to be decomposed. And the special noise is added to it. It is as shown in equation (2):

$$x^{(i)} = x + \beta_1 E_1(w^{(i)}) \quad (2)$$

in the formula (2), β_1 represents the expected signal-to-noise ratio of the first decomposition iteration. The first-order modal components $E_1(\bullet)$ is decomposed by the empirical mode decomposition (EMD) method. $w^{(i)}$ represents the i th group white noise used for auxiliary decomposition. The $x^{(i)}$ is the time series of WP after adding the special noise of the first group.

Step 2: The residual r_1 of the initial decomposition is calculated. And the modal component d_1 is obtained by the sequence of WP x minus the residual r_1 . They are as shown in equation (3) and equation (4):

$$r_1 = \frac{1}{I} \sum M(x + \beta_1 E_1(w^{(i)})) \quad (3)$$

$$d_1 = x - r_1 \quad (4)$$

where, the operator that is the local mean of the signal is generated by $M(\bullet)$. The number of groups of white noise is I .

Step 3: The signals of the white noise are continued to add. And the second residual r_2 is calculated. The second-order modal component d_2 of the original time series of WP. They are as shown in equation (4) and equation (5):

$$r_2 = \frac{1}{I} \sum M(r_1 + \beta_2 E_2(w^{(i)})) \quad (4)$$

$$d_2 = r_1 - r_2 \quad (5)$$

in the formula, β_1 represents the expected signal-to-noise ratio of the second decomposition. The second-order modal components $E_2(\cdot)$ is generated using the EMD method.

Step 4: It is Similar to Step 3. The k -th residual and k -th modal component are calculated. They are as shown in equation (6) and equation (7):

$$r_k = \frac{1}{I} \sum M(r_{k-1} + \beta_k E_k(w^{(i)})) \quad (6)$$

$$d_k = r_{k-1} - r_k \quad (7)$$

where, r_k represents k -th residual. d_k represents k -th modal component. r_{k-1} represents $(k-1)$ -th residual. β_k represents the expected signal-to-noise ratio of the k -th decomposition.

Step 5: the step 4 is repeated until the maximum number of iterations is reached or the residual cannot be further decomposed.

2.2.3. Permutation entropy

Entropy can be used to evaluate the complexity of time series, and entropy theory can be used to study the correlation between different time series. In order to avoid the complexity of WP prediction caused by the various sub-components obtained from ICEEMDAN decomposition, the permutation entropy (PE) is used to reconstruct the decomposed sub-components. PE, information entropy, fuzzy entropy, and other indicators are commonly used to evaluate the complexity of time series. When calculating the complexity between reconstructed subsequences, the concept of permutation is considered. The steps to calculate the PE of components for time series are as follows.

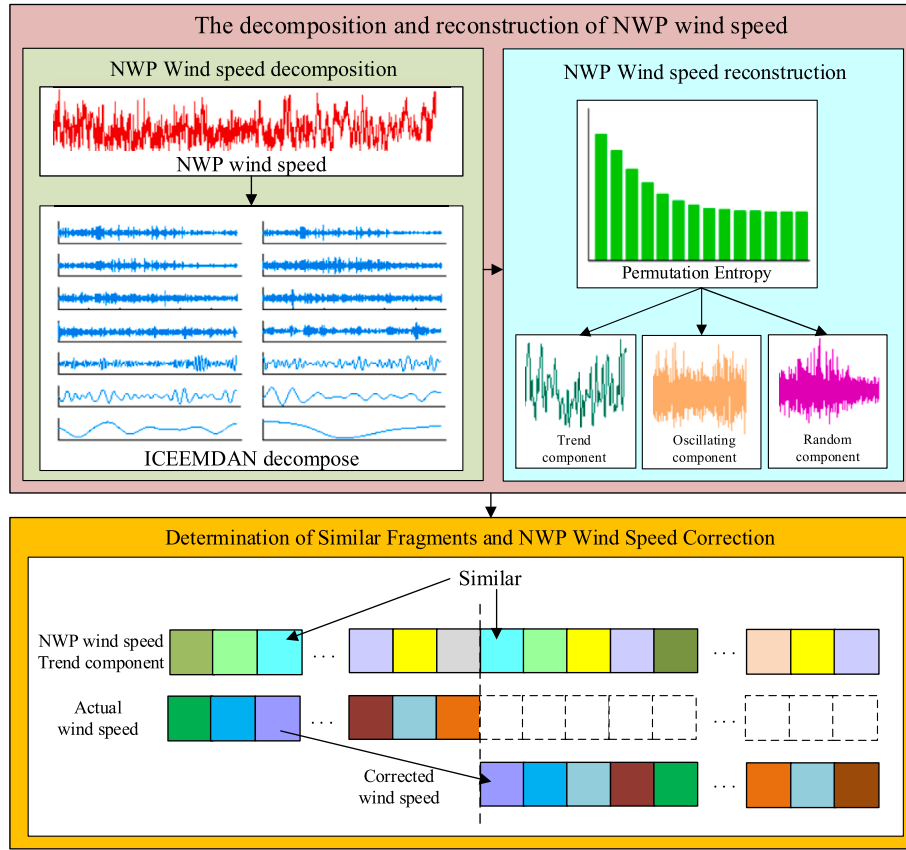


Fig. 4. Overall flowchart of NWP wind speed correction.

Step1: There is a time series $u(1), u(2), u(3), \dots, u(N)$ with a length of N . By reconstructing the original sequence, each subsequence $X(i)$ is represented. The process of reconstructed signal are as follows:

$$X(i) = u(i), u(i+L), \dots, u(i+(m-1)L) \quad (8)$$

where, m represents the number of bits embedded in the time series. L represents the delay of the time series.

Step2: The perform incremental sorting on each $X(i)$ internally, that is:

$$u(i+(j_1-1)L) \leq u(i+(j_2-1)L) \leq \dots \leq u(i+(j_m-1)L) \quad (9)$$

in this way, $X(i)$ is mapped to $(j_1, j_2, \dots, j_m)$. This is exactly one of the $m!$ arrangements. That is to say, every m -dimensional subsequence $X(i)$ is mapped to $m!$ One of the arrangements. The probability distribution of all time series is represented as $P_1, P_2, \dots, P_K$. Wherein, $K \leq m!$.

Step3: The Shannon Entropy is caculated. So, the PE of time series $u(1), u(2), u(3), \dots, u(N)$ is:

$$H(m) = - \sum_{j=1}^K P_j \ln P_j \quad (10)$$

where, $H(m)$ represents PE. When $P_j = 1/m!$, each symbol has and their probabilities are equal. At this time, the complexity of the time series is the highest. So, the PE that is $\ln(m!)$ is the highest.

Step4: In addition, for ease of representation, $H(m)$ is usually normalized:

$$G(m) = H(m)/\ln(m!) \quad (11)$$

where, $G(m)$ indicates the normalized PE.

The larger the PE, the stronger the randomness of the time series. Conversely, the weaker the randomness of the time series.

2.2.4. Reconstruction of wind speed sequence

To determine the variation trend of NWP wind speed, the PE corresponding to the sub-components of the wind speed sequence is used to be reconstructed random components, oscillatory components, and trend components for each sub-sequence of NWP wind speed. It is a key issue to determining the threshold for dividing random components, oscillatory components, and trend components. So, the Kmeans clustering algorithm is used to divide the PE of wind speed subsequences into three categories. The wind speed subsequences of each category are added together to obtain the random component, oscillation component, and trend component of the wind speed subsequence. The reconstructed trend component is carried to search for similar segments of wind speed sequences.

2.2.5. Error correlation ratio

The NWP wind speed in the future period is compared with the NWP wind speed in the past period. If the similarity between the two intervals, the NWP wind speed in the past period is inplaced with the NWP wind speed in the future period. So, the NWP wind speed is corrected. To avoid the influence of random components and oscillatory components of wind speed sequences on the similarity of wind speed segments, the trend components in NWP wind speed are used as the main data for searching for similar segments.

To quantify the similarity of NWP wind speeds between two time periods, the concept of ECR was proposed. The formula for calculating the ECR is shown in equation (12):

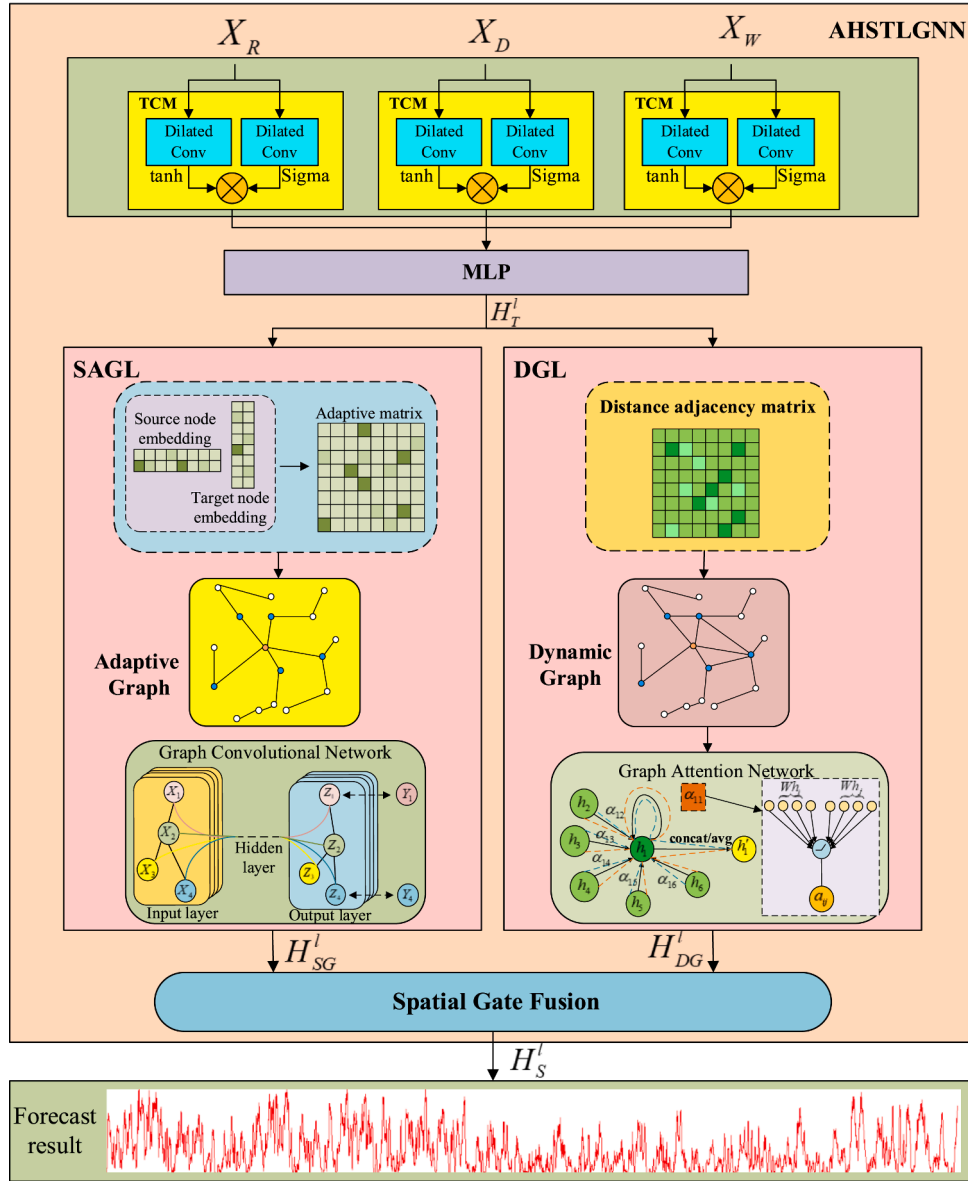


Fig. 5. Flow chart of short-term wind power cluster power prediction method.

$$ECR = \frac{RMSE}{PCC} (ECR > 0) \quad (12)$$

in the formula (21), root mean square error (RMSE) represents the root mean square error of two time periods, and pearson correlation coefficient (PCC) represents the pearson correlation coefficient of two time periods. The calculation methods for RMSE and PCC are shown in equations (13) and (14):

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (x_i - \hat{x}_i)^2}{n}} \quad (13)$$

$$PCC = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{Y})^2}} \quad (14)$$

in equations (13) and (14), x_i represents the value of NWP wind speed at historical time points, y_i represents the value of NWP wind speed at future time points. $\bar{X}$ represents the average NWP wind speed in his-

torical time periods, and $\bar{Y}$ represents the average NWP wind speed in future time periods.

RMSE reflects the data deviation between two time periods. And a smaller RMSE value indicates a smaller deviation of data between the two time periods. The PCC reflects the linear dependence of data between two time periods, and a larger PCC indicates a higher degree of linear dependence between the two data periods. In addition, when the PCC is greater than zero, the data from the two time periods are positively correlated. When the PCC is less than zero, the data from two time periods are negatively correlated. In summary, the smaller the value of the ECR, the higher the similarity of the data between the two time periods when the value of ECR.

2.2.6. The overall process of NWP wind speed correction

The overall process of the correction of NWP wind speed is as follows:

Step 1: The original NWP wind speed is decomposed by the ICE-EMDAN algorithm and the PE of each sub-component is calculated.

Step 2: The wind speed sub-components corresponding to the three categories are reconstructed into random components, oscillatory components, and trend components. And the trend components as the basis are used to selecting similar NWP wind speed segments in the future.

Step 3: the similar segments of wind speed can be obtained by calculating the ECR. If the ECR of two wind speed data segments is positive and the smaller the value, the higher the similarity between the two segments.

Step 4: the two sequence fragments that meet the requirements of wind speed are find. The actual wind speed corresponding to the NWP wind speed in the past period is replaced with the NWP wind speed in the future period to achieve the correction of NWP wind speed.

The overall flowchart of the correction NWP wind speed is shown in the Fig. 4.

2.3. Data input

To further explore the information of wind power, three periodic components were introduced to capture the dependence of multi-period time (Wang et al., 2023). Among them, T_R, T_D , and T_W respectively represent the historical time lengths corresponding to the historical time steps input into Temporal Convolution Module (TCM), with corresponding time lengths of 4 h, 1 day, and 1 week. And L_D, L_W represent the selected past days and past weeks respectively. Overall, a short-term wind power prediction is influenced by recent trends of wind power. Let the current time length be t and q be the sampling frequency per day. The characteristics of NWP for recent component extraction and prediction period is $X_R = [X_{t-T_R+1}, X_{t-T_R+2}, \dots, X_t] \in \mathbb{R}^{T_R \times N \times F}$ as the feature input of daily time series. The length of time is 4 h. The characteristics of wind power corresponding to the daily numerical weather forecast features is modeled. As for the L_d^{th} day, $X_d = [X_{t-L_d+1}, X_{t-L_d+2}, \dots, X_{t-L_d+T_D}] \in \mathbb{R}^{T_D \times N \times F}$ is the feature input as the daily time series, where $L_d \in \{1, 2, \dots, L_D\}$. The X_D is obtained by adding all X_d . The features of NWP are extracted in recent weeks. As for the L_w^{th} week, $X_W = [X_{t-7 \times L_w+1}, X_{t-7 \times L_w+2}, \dots, X_{t-7 \times L_w+T_W}] \in \mathbb{R}^{T_W \times N \times F}$, where $L_w \in \{1, 2, \dots, L_W\}$. In order to obtain the important features of the previous L_W week, the same input method as X_D is used to obtain X_W as the input for the week component. As shown in Fig. 5, H^{l-1} represents the inputs of the AHSTL block, which are the outputs of the $(l-1)^{th}$ TCM in block. The outputs of TCM and Adaptive Hybrid Graph Learning Module (AHGLM) in the first block are represented as H_T^l and H_S^l . To maintain consistency, the historical time step of the time series is represented as T .

2.4. Adaptive spatiotemporal graph framework

2.4.1. Time convolution module

To explore the complex temporal correlation and periodicity of wind power sequences, three time convolutional networks are combined to form a time convolutional module. The original inputs of the time convolution module are X_R, X_D and X_W . The time convolutional network has an exponential receptive field of extended causal convolution, and the gating mechanism can enhance the processing capabilities and modeling capabilities of time series. The following equation describes the operational relationship of a time convolutional network:

$$H^l = \tanh(W_{t1} * H^{l-1} + b_{t1}) \odot \sigma(W_{t2} * H^{l-1} + b_{t2}) \quad (15)$$

among them, $*$ represents the convolution operator. W_{t1}, W_{t2}, b_{t1} , and b_{t2} are learnable parameters. $\odot$ are element products. σ represent the sigmoid activation function, and $\tanh$ represents the activation function that is hyperbolic tangent. After processing, the time series is finally transmitted to a multilayer perceptron (MLP), and the final output of the

l -th TCM is $H_T^l \in \mathbb{R}^{T \times N \times D}$.

2.4.2. Adaptive hybrid graph learning module

AHGLM includes Static Adaptive Graph Learning (SAGL), Dynamic Graph Learning (DGL), and Space Gate Fusion (SGF). As shown in Fig. 2, the outputs of SAGL and DGL is represented as H_{SG}^l and H_{DG}^l . After passing through the spatial fusion gate, the output of the l -th AHGLM is H_S^l .

In SAGL, GCN is used to construct a topological graph network to represent the spatial relationships between wind farms and achieve spatial feature information fusion. According to reference (Kipf and Welling, 2017), graph convolution operations in the spectral domain can be approximated well by the first Chebyshev polynomial expansion and it can be extended to high-dimensional GCNs.

$$H = \left(I_N + D^{\frac{1}{2}} A D^{-\frac{1}{2}} \right) X \Theta \quad (16)$$

in the formula (16), A is the adjacency matrix and D is the degree matrix. $I_N = \mathbb{R}^{N \times N}$ is the identity matrix, and Θ is the graph convolution kernel.

Different wind farms in different geographical locations can be abstracted into high-dimensional space by representation learning nodes. Therefore, SAGL implements a learnable graph node embedding E_G , where d represents the dimension of node embedding. In addition, the adaptive adjacency matrix A_{adp} of the static graph will be obtained by multiplying E_G with E_G^T . The formula is as follows:

$$A_{adp} = \text{softmax}(\text{RELU}(E_G, E_G^T)) \quad (17)$$

among them, $H_{SG}^l = (I_N + A_{adp}) H_T^l \Theta$ is the output of SAGL, which $\Theta \in \mathbb{R}^{D \times D}$ is a learnable graph convolution kernel.

2.4.3. Dynamic graph learning

Dynamic graph learning (DGL) achieves the fusion of feature information in dynamic space through graph attention network (GAT). Because GAT can capture the dynamic impact of nearby wind farms. In addition, the decay matrix of exponential distance is used as the prior graph structure of GAT it is called as the distance adjacency matrix $A_{dis} \in \mathbb{R}^{N \times N}$. As shown in Fig. 2, GAT achieves the fusion of weighted feature by calculating the attention scores of the central node and neighboring nodes. In addition, multi-head attention is used to capture the relationship features between nodes from different learning sub-spaces. The complexity of time can also be reduced through a multi-head attention mechanism with parallel computing. The graph attention mechanism of DGL regarding graph nodes can be formalized as follows:

$$a_{ij} = \frac{\exp(\text{LR}(a^T [Wh_{t,i} || Wh_{t,j}]))}{\sum_{k \in N_i} \exp(\text{LR}(a^T [Wh_{t,i} || Wh_{t,k}]))} \quad (18)$$

$$\tilde{h}_{t,i} = \sigma \left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in N_i} a_{ij} W^k h_{t,j} \right) \quad (19)$$

among them, $a_{ij} \in \mathbb{R}$ represents the calculated attention score between v_i and v_j , where $j \in N_i$. N_i represents the set of neighbors of v_i . $\text{LR}(\bullet)$ represents LeakyReLU function. $a \in \mathbb{R}^{2D \times 1}$ is a learnable parameter vector. $W \in \mathbb{R}^{D \times D}$ is a weight matrix for shared linear transformations. $h_{t,i} \in \mathbb{R}^D$ represents the features of time in v_i to test t th step in H_T^l . $h_{t,j} \in \mathbb{R}^D$ represents the features of time in v_j to test t th step in H_T^l . $h_{t,k} \in \mathbb{R}^D$ represents the features of time in v_k to test t th step in H_T^l . wherein, v_i, v_j and v_k is the different point of topology map generated respectively. The features generated by multi-head attention are aggregated through addition. K represents the number of heads and $W^k \in \mathbb{R}^{D \times D}$ is the transformation parameter matrix corresponding to head k . $\tilde{h}_{t,i} \in \mathbb{R}^D$ is

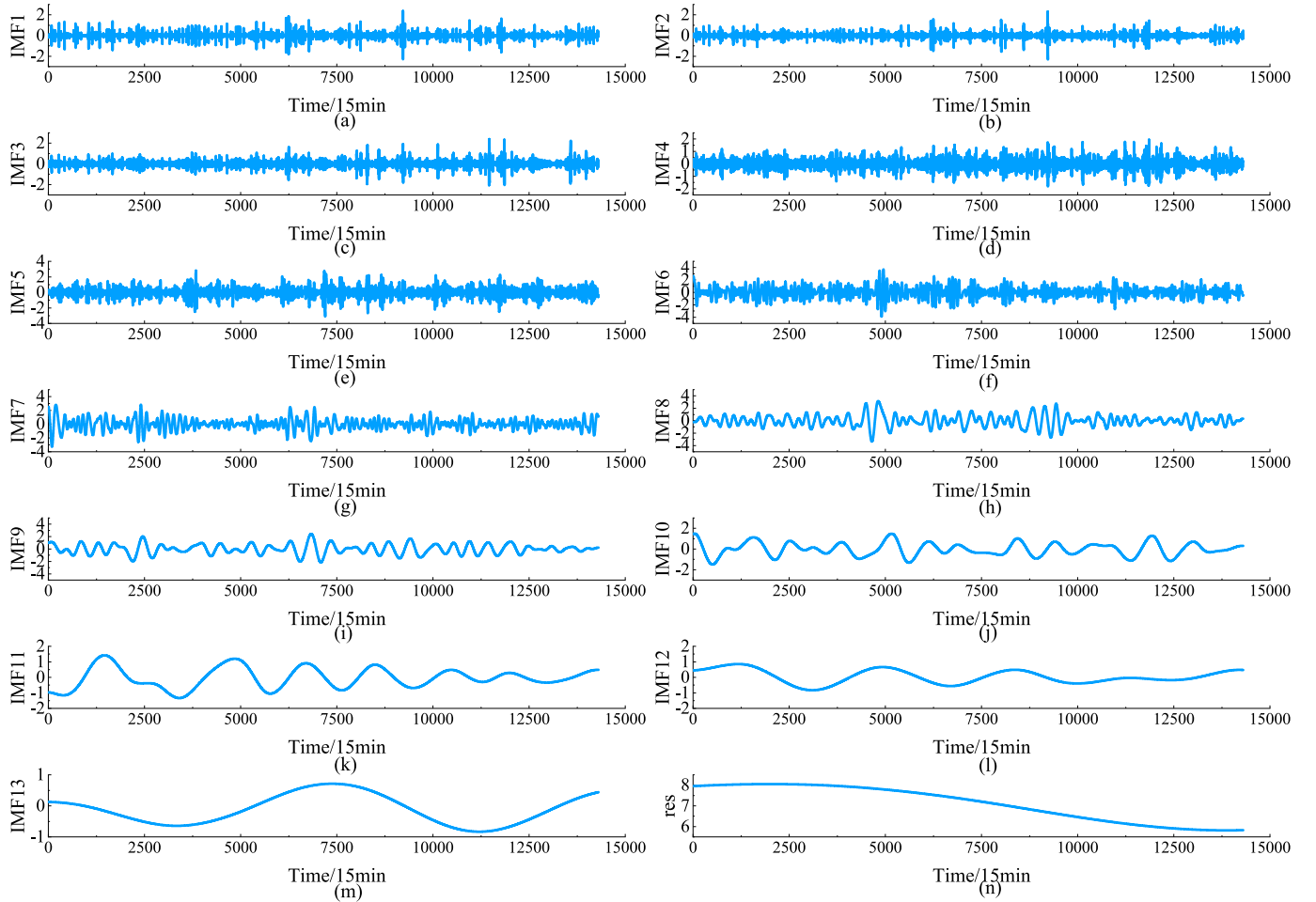


Fig. 6. Schematic diagram of the decomposition results of historical NWP wind speed.

the output of DGL at time step t . In the l -th block, the final output of DGL is $H_{DG}^l \in \mathbb{R}^{T \times N \times D}$. $\sigma(\cdot)$ indicates sigma function.

The two input features that are H_{SG}^l and H_{DG}^l are fused through spaces. The inputs come from the SAGL and DGL that mentioned above in the l -th block. As shown in Fig. 2, the gate fusion mechanism can be represented as:

$$H_S^l = \text{gate} \odot H_{SG}^l + (1 - \text{gate}) \odot H_{DG}^l \quad (20)$$

$$\text{gate} = \sigma(H_{SG}^l W_{g1} + H_{DG}^l W_{g2} + b_g) \quad (21)$$

among them, $W_{g1} \in \mathbb{R}^{D \times D}$, $W_{g2} \in \mathbb{R}^{D \times D}$ and $b_g \in \mathbb{R}^D$ are both learnable parameters. The final output of AHGLM is $H_S^l \in \mathbb{R}^{T \times N \times D}$. Space door fusion can adaptively control the static and dynamic data features of spatial correlation of each wind farm.

2.4.4. Output layer

For each output result H_S^l , a skip connection layer approach was used to directly connect them to the output layer. By stacking multiple adaptive hybrid spatial-temporal learning (AHSTL), the temporal receptive field of the structure of adaptive hybrid spatial-temporal graph neural network (AHSTGNN) was increased. Bottom layer can pay more attention to the temporal characteristics between adjacent WFs. The high-level is used to focus on long-term temporal information. And the modeling problems of spatial dependency from different time levels can be solved by skip connections. The outputs of all AHSTL module can be fused by summation. The result is $H_{out} \in \mathbb{R}^{T \times N \times D}$. Finally, two fully connected layers are used to implement the output layer to generate the

final results of WPP. The final output layer can be calculated by the following formula:

$$\hat{Y} = \text{RELU}(H_{out} W_{f1} + b_{f1}) \cdot W_{f2} + b_{f2} \quad (22)$$

wherein, $W_{f1} \in \mathbb{R}^{(TD) \times C}$, $W_{f2} \in \mathbb{R}^{C \times (MF)}$, $b_{f1} \in \mathbb{R}^C$ and $b_{f2} \in \mathbb{R}^{MF}$ are learnable parameter. $\hat{Y}$ is the final output result of the framework.

2.4.5. Loss function

The average absolute error between predicted and true values can be regressed as the loss function. And it can be minimized by back-propagation. The formula for the loss function is:

$$L(\theta) = \frac{1}{M} \sum_{i=t+1}^{t+M} |Y_i - \hat{Y}_i| \quad (23)$$

among them, θ represents all learnable parameters of our model. $\hat{Y}_i$ is the predicted power of the model for t a large-scale WFC at time step i . Y_i is the actual power.

2.5. Prediction error

This article uses root mean square error (RMSE), mean absolute error (MAE), coefficient of determination (R^2), and accuracy rate (AR) to verify the effectiveness of the proposed method. The calculation formulas for RMSE, MAE, R^2 , and AR are shown in the formula (24) ~ (27):

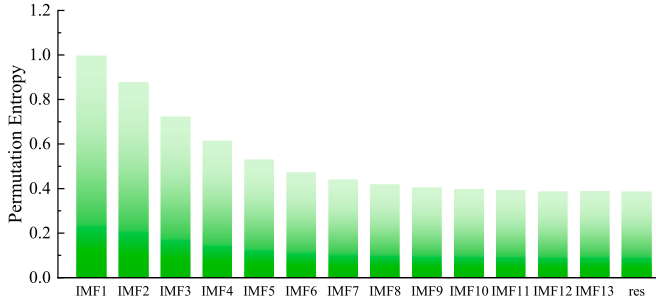


Fig. 7. Schematic diagram of permutation entropy of historical NWP wind speed sub components.

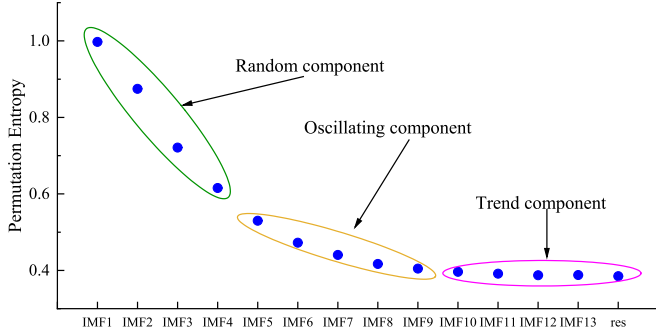


Fig. 8. Schematic diagram of the distribution of random components, oscillatory components, and trend components.

$$R_{mse} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{nCap^2}} \quad (24)$$

$$R_{MAE} = \frac{1}{n} \sum_{k=1}^n \left(\frac{|y_i - \hat{y}_i|}{Cap} \right) \quad (25)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (26)$$

$$R_{AR} = 1 - \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{nCap^2}} \quad (27)$$

in Equation (24), Equation (25), Equation (26) and Equation (27), where y_i represents the actual power at time i , $\hat{y}_i$ represents the predicted power at time i , $\bar{y}$ represents the average of the actual power at time i , n represents the length of the test set, and Cap represents the actual start-up capacity of WF. The installed capacity in this paper is used because the actual start-up capacity of WF is difficult to obtain.

2.6. The process of the short-term power prediction for wind farm cluster

- (1) Firstly, the method of NWP wind speed correction in Section 2.2 is used to correct the NWP wind speed of each wind farm. Then, the corrected NWP wind speed, wind direction, temperature, and humidity characteristics are input into the time convolution module. Next, the data features are output through MLP;
- (2) The Establishment of adjacency matrix. The adjacency matrix is constructed based on the information of historical actual wind speed and geographic location, which serves as the foundation matrix for adaptive spatiotemporal graph network for prediction.
- (3) The output of data features is divided into two categories by the MLP. The first category used to construct an adaptive matrix for

constructing an adaptive topology graph. The second category of data is used to construct distance adjacency matrices for constructing dynamic topological graphs.

- (4) The adaptive matrix and dynamic adjacency matrix are input into the graph convolutional neural network and graph attention neural network respectively. then the output of data features is fused. Finally, the predicted power can be obtained for WFC.

3. Example analysis

To verify the effectiveness of the proposed method, simulation experiments were conducted using data from two large-scale WFCs in China. Two WFCs consist of 50 wind farms with installed capacities of 6855.70 MW and 4413.85 MW, respectively. Two WFCs are represented by WFC A and WFC B.

3.1. NWP wind speed correction

To prevent the random components in the original NWP wind speed from affecting the correction process, it is necessary to capture the trend components in the original NWP wind speed. Firstly, we take the first wind farm of WFC A as an example to describe process of the NWP wind speed correction for wind farm.

3.1.1. Decomposition of wind speed

The original historical NWP wind speed of the first wind farm in WFC A was decomposed using the ICEEMDAN algorithm. The decomposition results are shown in Fig. 6. The historical predicted wind speed was decomposed into 13 sub-components and 1 residual component, which includes high-frequency, mid frequency, and low-frequency components. To avoid computational redundancy, the PE of the above 14 components was calculated separately. And the above components were reconstructed into random components, oscillatory components, and trend components according to the principle of similar PE. The PE calculated based on the decomposed components is shown in Fig. 7.

As shown in Fig. 7, the values of PE decrease sequentially with the increase of component names. And the values of PE between adjacent names also decrease sequentially. Although the residual component of historical NWP wind speed can reflect the trend changes of historical wind speed, the smoothness of this curve is high and cannot represent the fluctuation information of historical wind speed. Therefore, reconstructing the trend component can quantify the trend information and fluctuation information of historical wind speed.

The important indicator for dividing PE is to determine the threshold between random components, oscillatory components, and trend components. This article firstly calculates the errors between the entropies of adjacent permutations, and then uses the k-means clustering algorithm to cluster the errors. At the same time, the wind speeds corresponding to the errors are clustered to obtain reconstructed random components, oscillatory components, and trend components. Fig. 8 shows the schematic diagram of the sub-component names corresponding to the random component, oscillation component, and trend component. It can be seen from the figure that IMF1 ~ IMF4 are determined as the random component interval, IMF5 ~ IMF9 are determined as the oscillation component interval, and the other components are determined as the trend component. Next, the components of the corresponding component are summed to obtain the reconstructed historical sequences of NWP wind speed. The reconstructed sequences of historical NWP wind speed are shown in Fig. 9.

From Fig. 9 (a), it can be seen that the characteristics of the random component are high oscillation frequency in terms of historical NWP wind speed and small average amplitude. According to Fig. 9 (b), the frequency of the oscillating component is lower than that of the random component. And the average amplitude of the oscillating component is higher than that of the random component. From Fig. 9 (c), it can be seen that the frequency of the trend component is the smallest compared

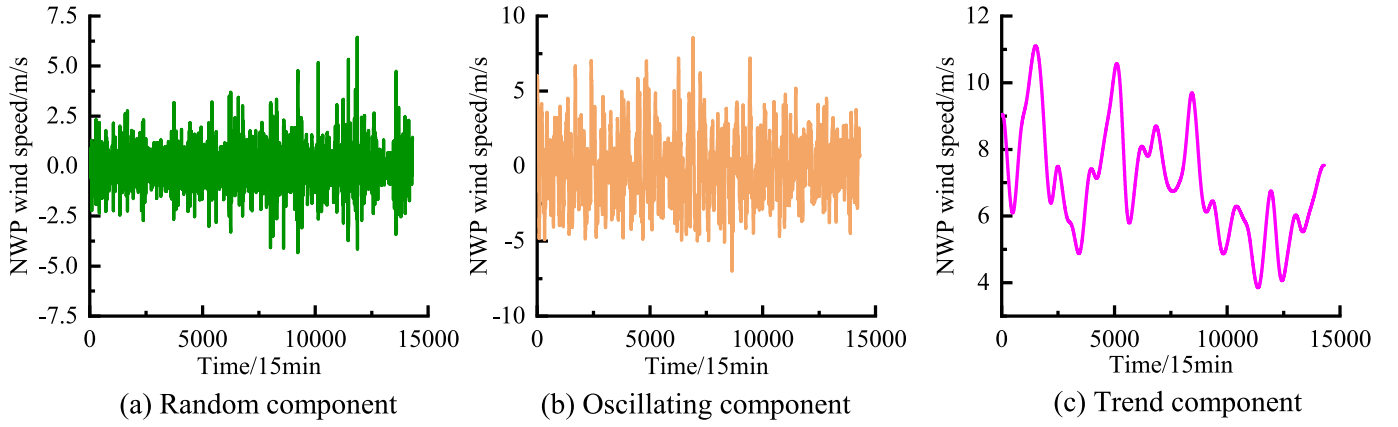


Fig. 9. Reconstructed Historical NWP Wind Speed Sequence.

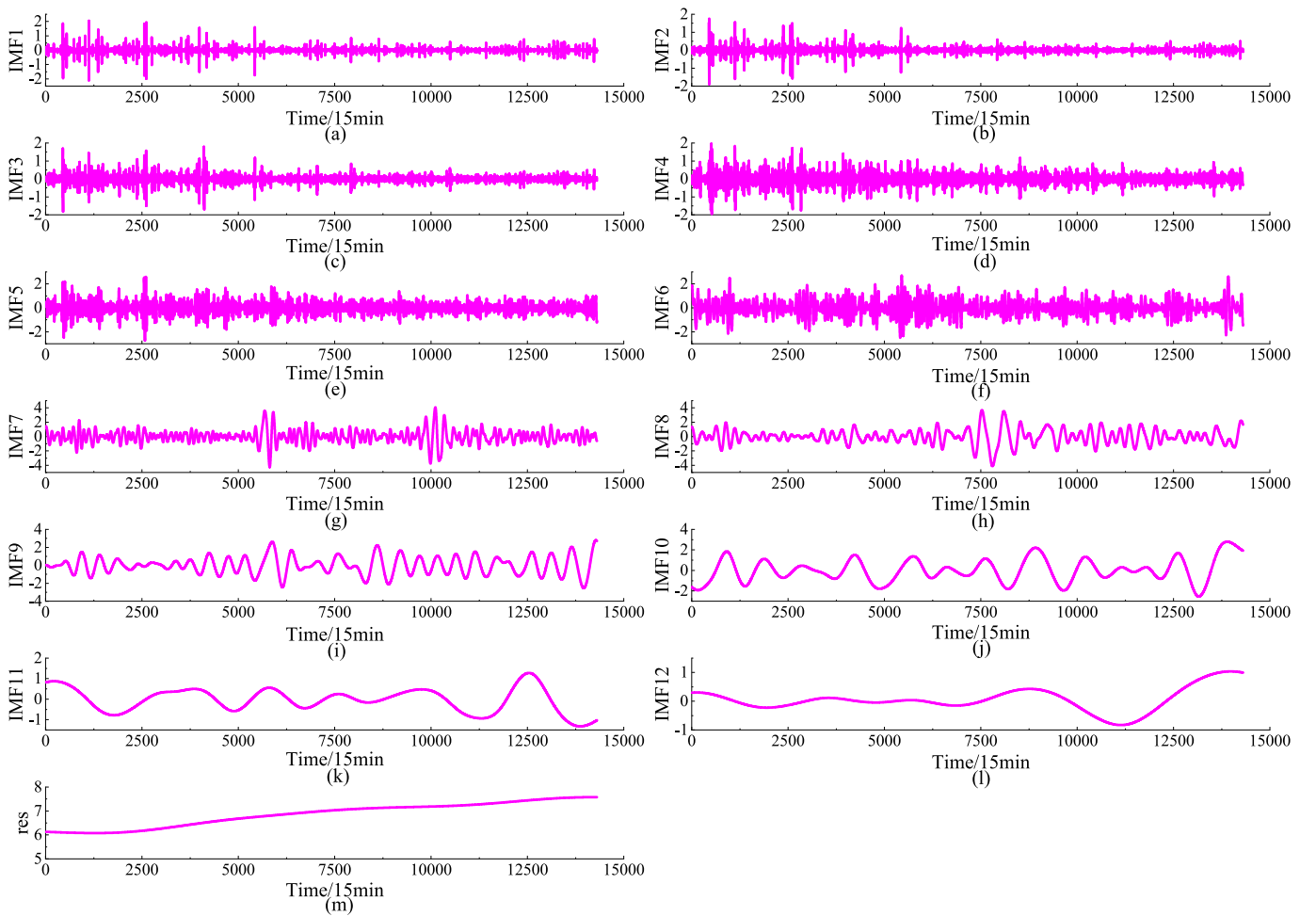


Fig. 10. Schematic diagram of future NWP wind speed decomposition.

to the random components and oscillatory components. The trend of historical NWP wind speed can be clearly observed. The amplitude of the trend component is between $[2.8 \sim 11.2]$. Comparing with the trend components and oscillatory components, the amplitude range of the trend component is the largest. In addition, the trend component of reconstructed historical NWP can not only reflect trend changes, but also describe the fluctuation characteristics of wind speed.

At the same time, the original future NWP wind speed of the first wind farm in WFC A was decomposed by the ICEEMDAN algorithm. The

decomposition results are shown in Fig. 10, which decomposes the historical predicted wind speed into 12 sub-components and 1 residual component. The decomposed components include high-frequency components, mid-frequency components, and low-frequency components. To avoid computational redundancy, the PE of the above 13 components was calculated separately. And the above components were reconstructed into random components, oscillatory components, and trend components according to the principle of similar PE. The PE calculated based on the decomposed components is shown in Fig. 11.

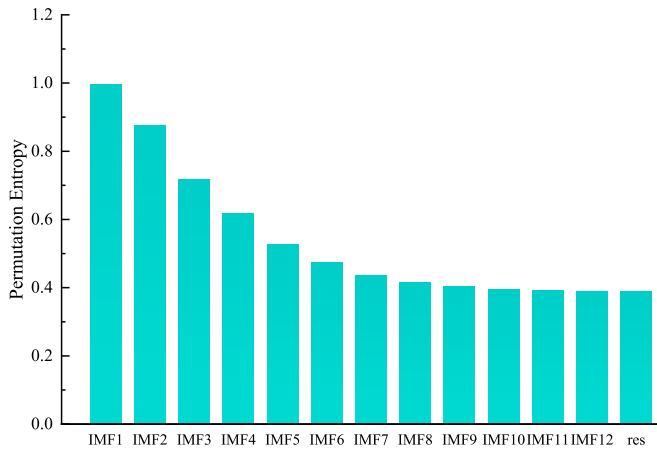


Fig. 11. Schematic diagram of permutation entropy of wind speed sub components in future NWP.

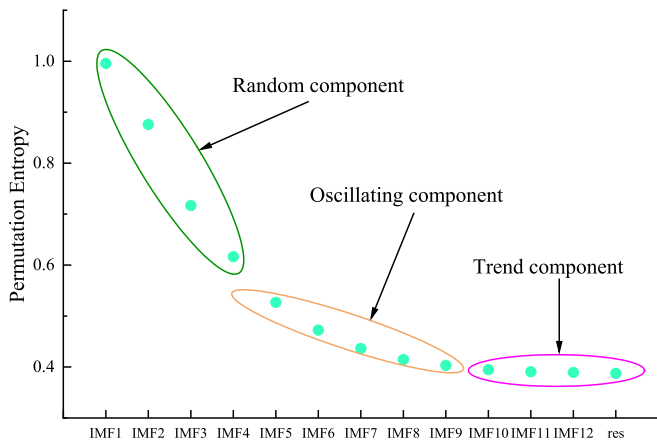


Fig. 12. Schematic diagram of the distribution of random, oscillatory, and trend components of future NWP wind speed.

As shown in Fig. 11, the value of PE decreases sequentially with the increase of component names. And the PE interpolation between adjacent names also decreases sequentially. Although the residual component of future NWP wind speed can reflect the trend changes of future wind speed, the smoothness of this curve is high and cannot represent the fluctuation information of future wind speed. Therefore, the reconstructed trend component quantifies the trend information and fluctuation information of future wind speed.

The important indicator for dividing PE is to determine the threshold between random components, oscillatory components, and trend

components. This article first calculates the error between the entropies of adjacent permutations. And then uses the k-means clustering algorithm to cluster the errors. At the same time, the wind speeds corresponding to the errors are clustered to obtain reconstructed random components, oscillatory components, and trend components. Fig. 12 shows the schematic diagram of the sub-component names corresponding to the random component, oscillation component, and trend component. It can be seen from the figure that IMF1 ~ IMF4 are determined as the random component interval, IMF5 ~ IMF9 are determined as the oscillation component interval, and the other component intervals are determined as the trend component interval. Next, the components of the corresponding component intervals are summed to obtain the reconstructed sequence of future NWP wind speed. The reconstructed sequences of historical NWP wind speed are shown in Fig. 13.

From Fig. 13 (a), it can be seen that in terms of future NWP wind speed, the characteristics of the random component are high oscillation frequency and small average amplitude. According to Fig. 13 (b), the frequency of the oscillating component is lower than that of the random component. And the average amplitude of the oscillating component is higher than that of the random component. From Fig. 13 (c), it can be seen that the frequency of the trend component is the smallest compared to the random components and oscillatory components. And the trend of historical NWP wind speed can be clearly observed. The amplitude of the trend component is between [3.8 ~ 10.8]. And comparing with the trend components and oscillatory components, the amplitude range of the trend component is the largest. In addition, the trend component of reconstructed historical NWP can not only reflect trend changes, but also describe the fluctuation characteristics of wind speed.

Fig. 14 depicts the Pearson correlation coefficient and the heatmap of root mean square error heatmap of historical NWP wind speeds and future NWP wind speeds at different time periods. According to 14 (a), the Pearson correlation coefficients of NWP wind speed in historical periods and future periods are not the same and have significant differences. There are both strong and positive correlations, as well as negative and uncorrelated relationships between each period. The Pearson correlation coefficient (purple part) of some time periods is close to -1 , indicating a negative correlation between the NWP wind speeds of two time periods. The Pearson correlation coefficient (green part) for some time periods is close to 0, showing weak or no correlation. The Pearson correlation coefficient (red part) for some time periods is close to 1, showing a strong correlation.

The Pearson correlation coefficient only describes the linear correlation between NWP wind speeds in two time periods. However, the quantitative relationship between NWP wind speeds cannot be measured in two time periods. Therefore, an indicator is still needed to measure the quantity of two time series. From Fig. 14 (b), it can be seen that the RMSE of NWP wind speed in different historical periods and future periods are not the same, but they have certain similarities.

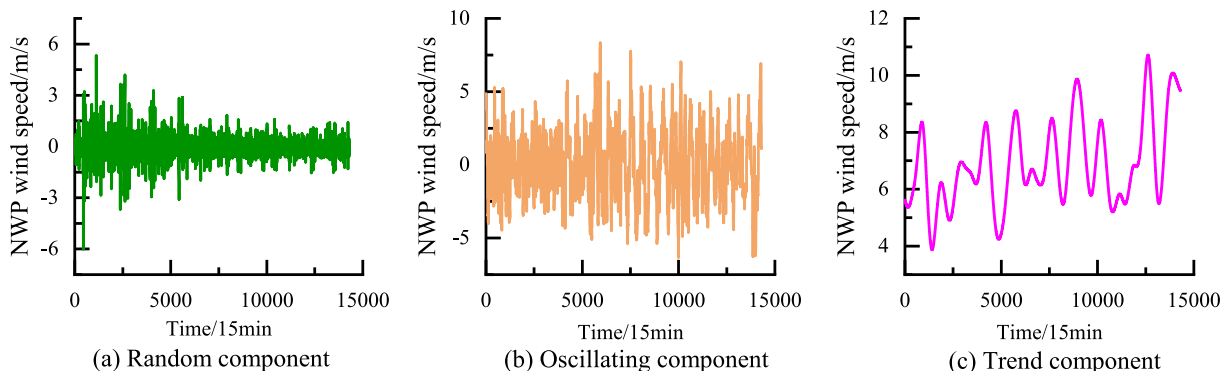


Fig. 13. Reconstructed Future NWP Wind Speed Sequence.

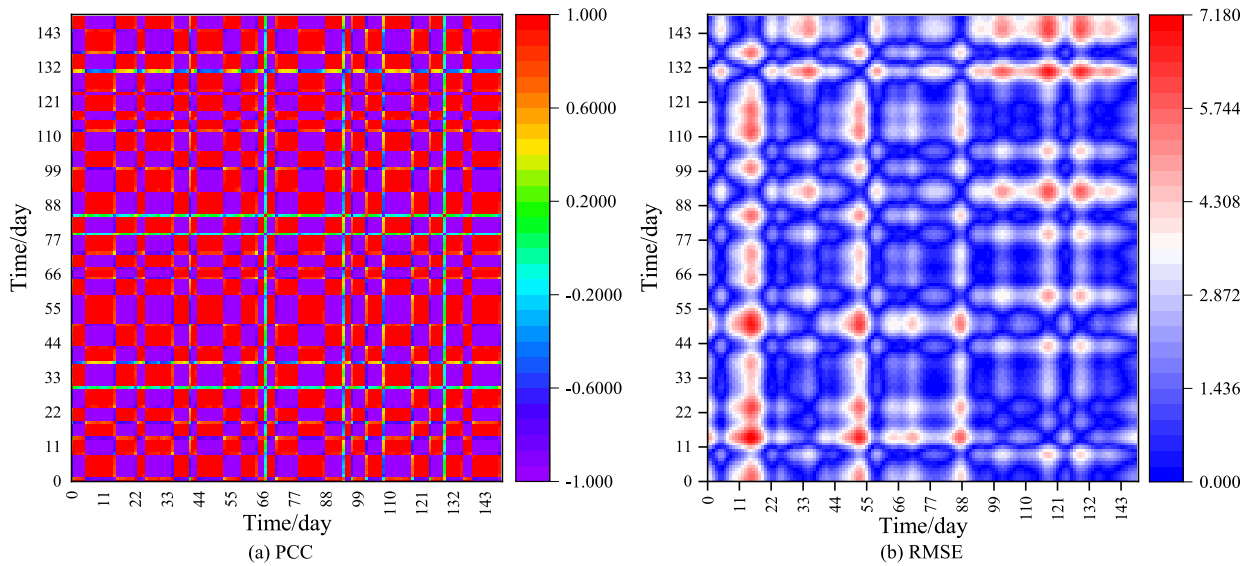


Fig. 14. The heatmap of historical and future NWP wind speeds for PCC and RMSE at different time periods.

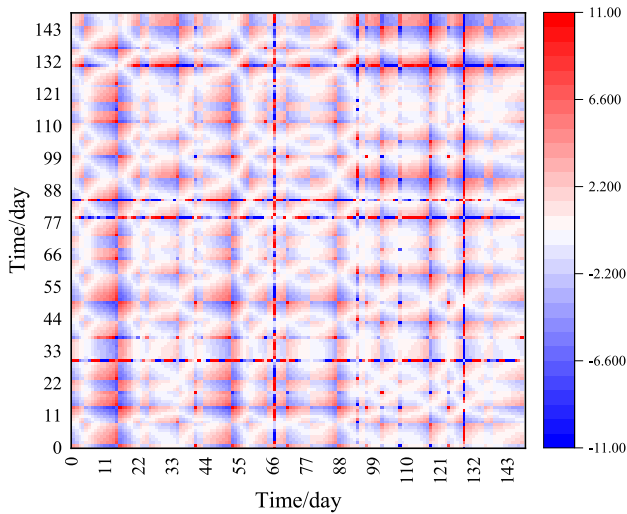


Fig. 15. Error correlation heatmap of past and future NWP wind speeds at different time periods.

Table 1
Numerical statistics of ECR.

Evaluating indicator	Maximum	Minimum	Average
ECR	192928.4	-52669.1	4.7036

Among them, the RMSE of the vast majority of time periods is less than 1.436 m/s, and the RMSE of a small number of time periods is greater than 5.744 m/s.

The ECR of each time period can be obtained by the RMSE of each time period divides the PCC of the corresponding time period. Fig. 15 shows the heatmap of the ECR of each time period. As shown in Fig. 15, it can be seen that there is a large difference in the ECR between future NWP wind speed and historical NWP wind speed. To ensure the convenience of plotting, we have processed some time periods with high ECRs. The number that the ECR is greater than 10 is set as 11. And the number that the ECR is less than -10 is set as -11. The specific values of the ECR are shown in Table 1.

According to Table 1, the maximum value of ECR is 192928.4, the

minimum value is -52669.1, and the average value is only 4.7036. It indicates that there are some segments that are not similar. Between the historical NWP wind speed and future NWP wind speed of the wind farm that are very dissimilar. However, these time segments are minority and the vast majority of ECRs are distributed around 4.7. It shows that there is a certain error relationship between the historical NWP wind speed and future NWP wind speed of the wind farm.

In order to measure the reliability of ECR, the time periods corresponding to different numbers of ECR are presented, and the comparison figures of historical periods and future periods under different numbers of ECR are drawn as shown in Fig. 15. RMSE displays the root mean square error of two time series, and the smaller the RMSE is, the smaller the error between the two time series is. The linear dependence between two time series is represented by PCC. When the PCC of two time series is positive, it indicates that the overall trends of the two time series are the same. When the PCC is negative, the overall trends of the two time series are opposite. The emergence of PCC effectively avoids the situation where the RMSE values of two time series are relatively small but the overall trend is opposite. ECR is the ratio of RMSE to PCC. When the RMSE is smaller and the PCC is closer to 1, the value of ECR is smaller. So, the two time series are more similar.

From Fig. 16, it can be seen that the fluctuation patterns of actual wind speed and predicted wind speed in Fig. 16 (a), (b), (c) and (f) are most similar. Furthermore, if ECR is positive, the overall trend of actual wind speed and predicted wind speed is consistent. If ECR is negative, it indicates that the overall trend of actual wind speed and predicted wind speed is opposite. Next, the NWP wind speed of all wind farms in WFC A and WFC B will be corrected using the method proposed in section 2.2. And WP prediction will be carried out using the method proposed in the article.

3.2. Wind power prediction

3.2.1. Wind power prediction for wind farm cluster A

The time range of data for the large-scale WFC A is from March 1, 2020 to December 30, 2020. The information of NWP includes wind speed, wind direction, temperature, and humidity. The time resolution of WP and NWP is both 15 min. The information of NWP is input into the AHSTGNN model for training. And the proportions of the training set, validation set, and test set are 70 %, 10 %, and 20 %, respectively. The scale of input time of the AHSTGNN model is 24 h, and the scale of output time is also 24 h. According to reference (Yang et al., 2025), we use grid optimization combined with ten-fold cross validation to

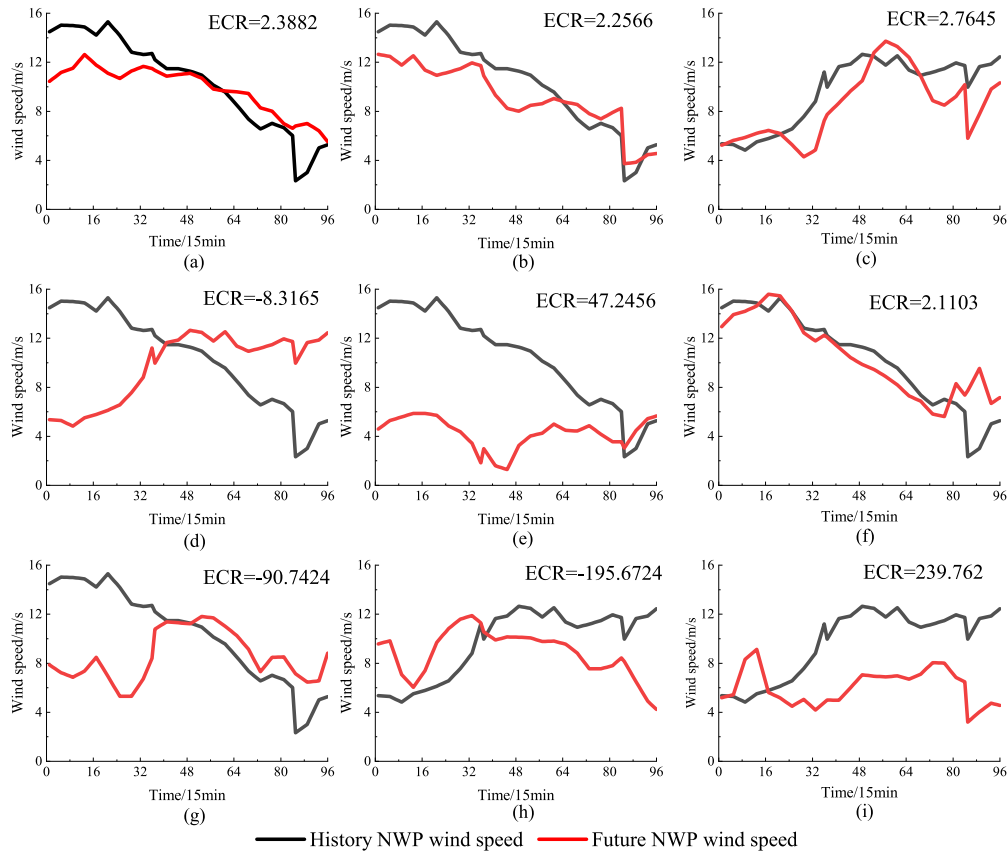


Fig. 16. Comparison of history NWP wind speeds → Comparison of history NWP wind speeds.

Table 2

Important hyperparameters of different prediction models for wind farm cluster A.

Model	Batch_size	Learn rate	Dropout	Epoch	Hidden_size
CNNLSTM	64	0.001	0.18	60	80
TCN	128	0.06	0.21	50	80
GRU	128	0.02	0.16	60	70
GNN	128	0.05	0.15	60	110
GCN	128	0.06	0.22	60	80
SGNN	64	0.005	0.15	50	90
MLP-WOA	128	0.001	0.20	60	90
LSTM-GOW	128	0.006	0.15	60	100
STGNN	64	0.001	0.15	50	128
AHSTGNN	64	0.001	0.15	50	128

optimize the hyperparameters of each model in the training set, making the model have robust predictive performance and generalization ability. The important hyperparameters for the different prediction models are shown in Table 2. for wind farm cluster A.

To verify the effectiveness of the method proposed in this paper, it was compared with CNNLSTM, TCN, GRU, GNN, GCN, SGNN, MLP-WOA, LSTM-GOW, STGNN, AHSTGNN, GNN-CN, GCN-CN, and STGNN-CN. The predicted results and actual power values of 14 models are shown in the Fig. 17. After calculation, the RMSE, MAE, R^2 , and accuracy rate of the 14 models are shown in Table 3 and Fig. 18. The Table 3 presents three types of comparisons such as ablation experiments, comparisons between various models, and comparisons of prediction accuracy before wind speed correction and after NWP wind speed correction.

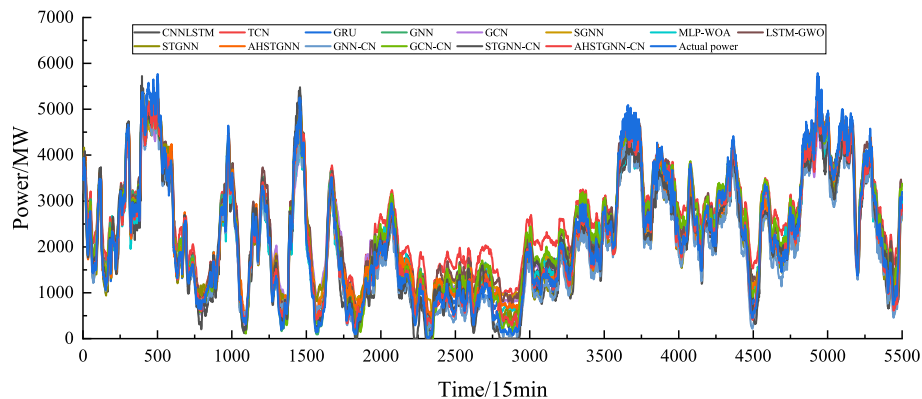
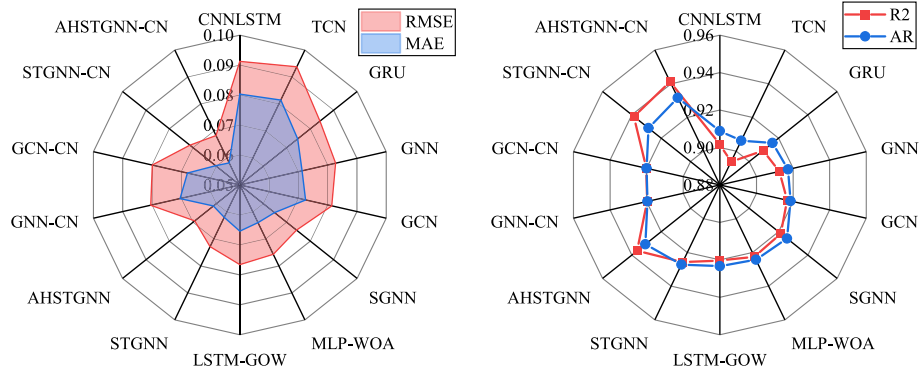


Fig. 17. Comparison of results of various prediction models.

Table 3

Evaluating indicator of comparing with other methods for wind farm cluster A.

Algorithm	RMSE	MAE	R ²	AR	Algorithm	RMSE	MAE	R ²	AR
CNNLSTM	0.09124	0.08017	0.8754	90.88 %	LSTM-GOW	0.07657	0.06524	0.9014	92.34 %
TCN	0.09376	0.08264	0.8718	90.62 %	STGNN	0.07285	0.06147	0.9127	92.72 %
GRU	0.08413	0.07292	0.8722	91.59 %	AHSTGNN	0.06923	0.05846	0.9345	93.08 %
GNN	0.08269	0.07141	0.8769	91.73 %	GNN-CN	0.08043	0.06927	0.8911	91.96 %
GCN	0.08137	0.07064	0.8834	91.86 %	GCN-CN	0.07987	0.06839	0.8945	92.01 %
SGNN	0.07416	0.06342	0.9083	92.58 %	STGNN-CN	0.07126	0.06056	0.9281	92.87 %
MLP-WOA	0.07569	0.06438	0.9052	92.43 %	AHSTGNN-CN	0.06828	0.05734	0.9412	93.17 %

**Fig. 18.** Comparison between different models for wind farm cluster A.**Table 4**

Important hyperparameters of AHSTGNN model wind farm cluster B.

Model	Batch_size	Learn rate	Dropout	Epoch	Hidden_size
CNNLSTM	64	0.005	0.21	80	90
TCN	64	0.008	0.16	60	80
GRU	128	0.04	0.18	60	80
GNN	128	0.06	0.20	60	100
GCN	128	0.08	0.17	80	85
SGNN	64	0.005	0.15	50	90
MLP-WOA	128	0.002	0.16	70	90
LSTM-GOW	128	0.007	0.19	60	100
STGNN	64	0.001	0.15	75	120
AHSTGNN	64	0.0001	0.15	50	128

(1) Ablation experiment

The algorithms involved in this section include AHSTGNN, STGNN, SGNN, and GCN. After comparison, GCN has the lowest prediction accuracy among the methods in this section. SGNN adds spatial attention mechanism compared to GCN, which can capture spatial correlation

features between wind farms. Compared to GCN, RMSE and MAE are reduced by 0.00721 and 0.00722, respectively, and R^2 and AR are improved by 0.0038 and 0.72 %, respectively. Compared with SGNN, STGNN adds a time attention mechanism that can capture the temporal correlation of internal feature information of wind farms, further improving the prediction accuracy of WP. Compared with SGNN, STGNN algorithm reduces RMSE and MAE by 0.00131 and 0.00195, and improves R^2 and AR by 0.0044 and 0.14 %, respectively. AHSTGNN has added adaptive dynamic adjustment function comparing with STGNN, which can adaptively match the changes in the topology of WFCs. After comparing the prediction accuracy of AHSTGNN with STGNN, the RMSE and MAE decreased by 0.00362 and 0.00301 respectively, and the R^2 and AR increased by 0.0218 and 0.36 % respectively. The above evaluation indicators further demonstrate the effectiveness of the proposed model in this paper.

(2) Comparison between various algorithms

The algorithms involved in this section include CNNLSTM, TCN, GRU, GNN, MLP-WOA, LSTM-WOA, and AHSTGNN. After comparison,

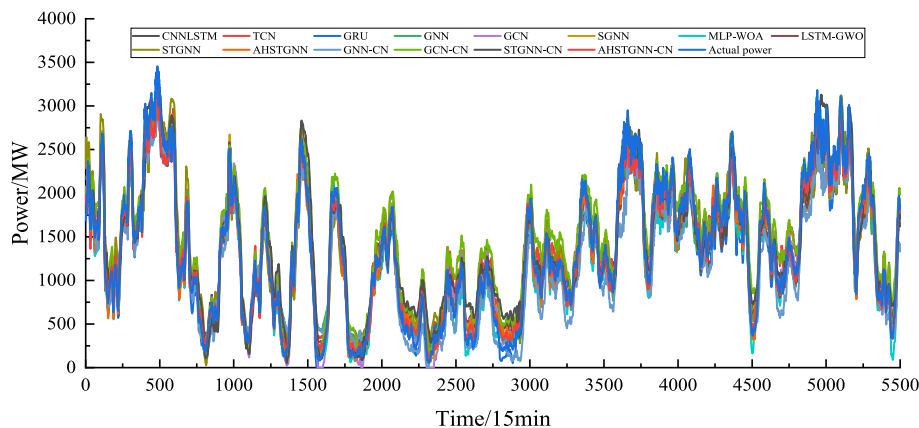
**Fig. 19.** Comparison of results of various prediction models.

Table 5

Evaluating indicator of comparing with other methods for wind farm cluster B.

Algorithm	RMSE	MAE	R ²	AR	Algorithm	RMSE	MAE	R ²	AR
CNNLSTM	0.08976	0.07821	0.8628	91.02 %	LSTM-GOW	0.07583	0.06438	0.8914	92.42 %
TCN	0.09145	0.08014	0.8642	90.86 %	STGNN	0.07395	0.06456	0.9034	92.61 %
GRU	0.08348	0.07268	0.8641	91.65 %	AHSTGNN	0.07314	0.06286	0.9234	92.69 %
GNN	0.08353	0.07231	0.8672	91.65 %	GNN-CN	0.07637	0.06520	0.8796	92.36 %
GCN	0.08219	0.07137	0.8735	91.78 %	GCN-CN	0.07621	0.06534	0.8823	92.38 %
SGNN	0.07435	0.06358	0.9014	92.57 %	STGNN-CN	0.07317	0.06186	0.9183	92.68 %
MLP-WOA	0.07504	0.06451	0.8945	92.50 %	AHSTGNN-CN	0.07247	0.06122	0.9276	92.75 %

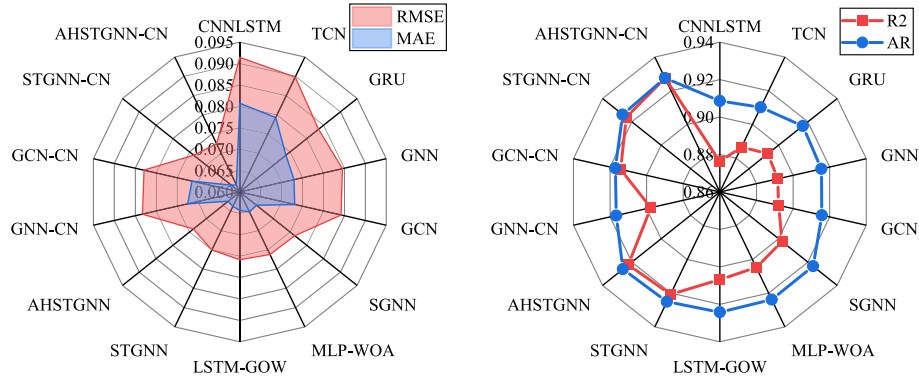


Fig. 20. Comparison between different models for wind farm cluster B.

the RMSE of the method proposed in this paper decreased by 0.02201, 0.02453, 0.01490, 0.01346, 0.00646, and 0.00734 compared to the other six methods, and the MAE decreased by 0.02171, 0.02418, 0.01446, 0.01295, 0.00592, and 0.00678 compared to the other six methods. The R² improved by 0.0591, 0.0627, 0.0623, 0.0576, 0.0293, and 0.0331 compared to the other six methods, and the accuracy improved by 2.20 %, 2.46 %, 1.49 %, and 1.39 % compared to the other six methods. 5 %, 0.65 %, and 0.74 %. Based on the above analysis, the model proposed in this article is superior to the compared algorithms,

demonstrating the superiority of the method proposed in this article.

(3) Comparison of prediction accuracy after NWP correction

The algorithms involved in this section include AHSTGNN and

Table 6

The average changes in evaluation indicators during ablation experiments.

Algorithm	RMSE↓	MAE↓	R ² ↑	AR↑
GCN	0.0106	0.01036	0.0505	1.07 %
SGNN	0.00307	0.00286	0.0241	0.31 %
STGNN	0.00226	0.00236	0.0209	0.22 %

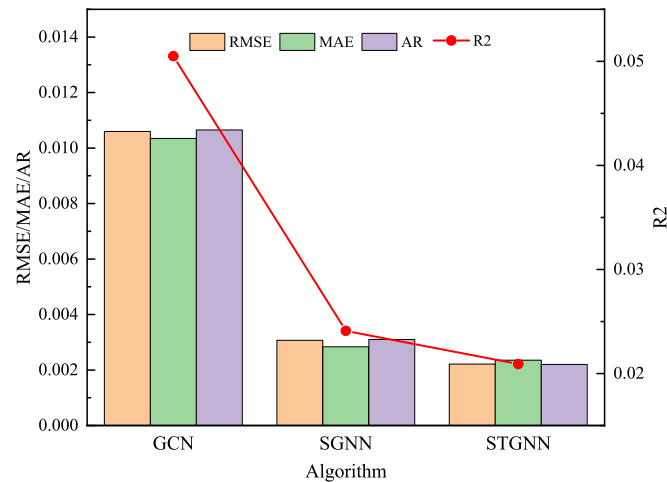


Fig. 21. Statistical chart of average changes in evaluation indicators for predicted results in ablation experiments.

Table 7

The average changes in evaluation indicators during ablation experiments.

Algorithm	RMSE↓	MAE↓	R ² ↑	AR↑
CNNLSTM	0.01931	0.01853	0.0599	1.94 %
TCN	0.02142	0.02073	0.0609	2.15 %
GRU	0.01262	0.01214	0.0608	1.27 %
GNN	0.01193	0.01120	0.0569	1.20 %
MLP-WOA	0.00418	0.00379	0.0291	0.42 %
LSTM-GOW	0.00502	0.00415	0.0326	0.51 %

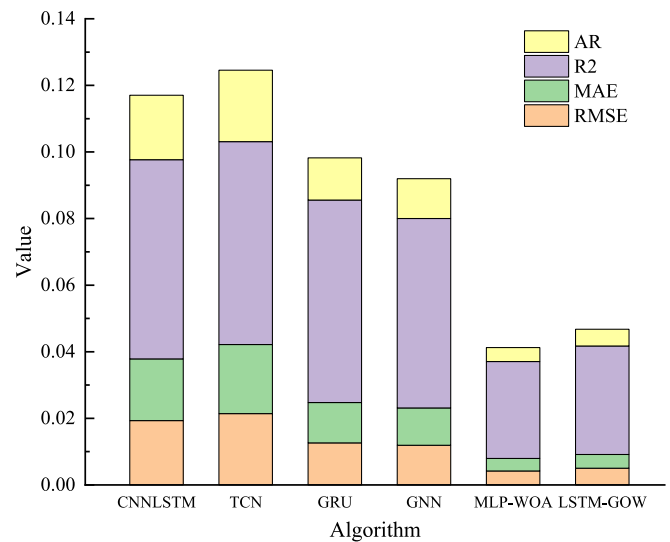
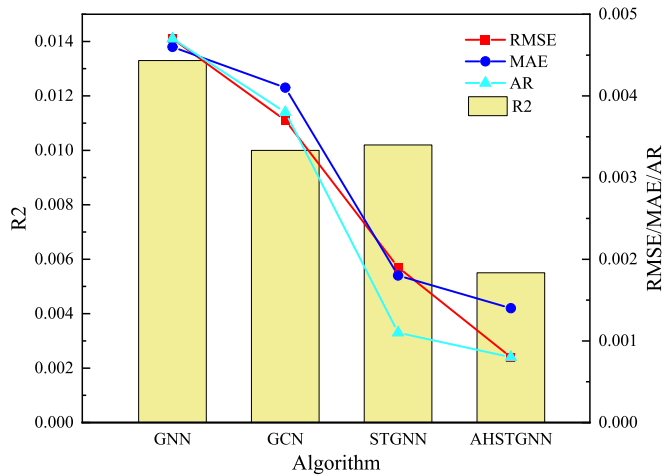


Fig. 22. Statistical chart of average changes in evaluation indicators for predicted results for different deep learning algorithms.

Table 8

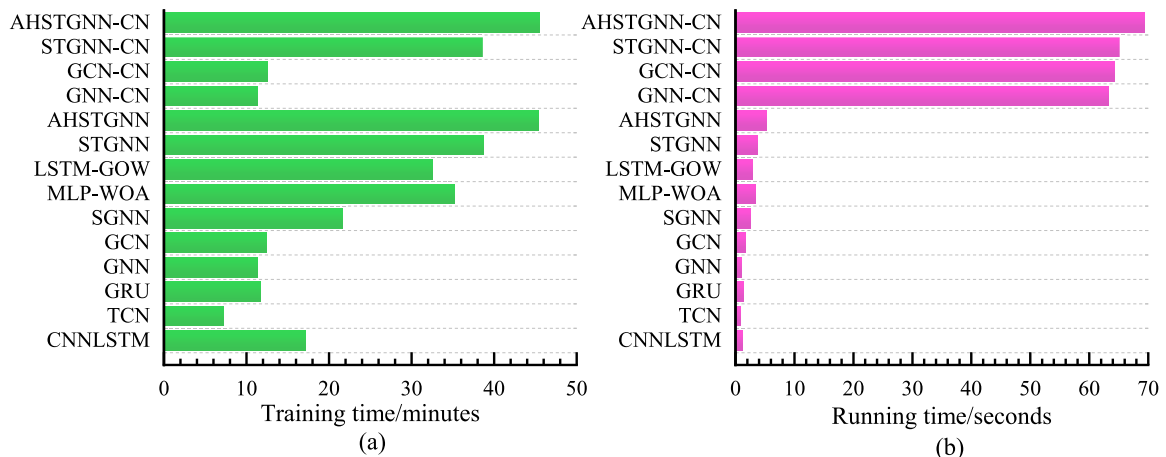
The average changes in evaluation indicators during ablation experiments.

Algorithm	RMSE↓	MAE↓	R ² ↑	AR↑
GNN	0.0047	0.0046	0.0133	0.47 %
GCN	0.0037	0.0041	0.0100	0.38 %
STGNN	0.0019	0.0018	0.0102	0.11 %
AHSTGNN	0.0008	0.0014	0.0055	0.08 %

**Fig. 23.** Statistical chart of average changes in evaluation indicators for predicted results after NWP correction.**Table 9**

Average training and running time of each model.

Algorithm	Training time (minutes)	Running time (seconds)	Algorithm	Training time (minutes)	Running time (seconds)
CNNLSTM	17.147	1.274	LSTM-GOW	32.475	2.954
TCN	7.247	0.851	STGNN	38.646	3.641
GRU	11.673	1.375	AHSTGNN	45.347	5.237
GNN	11.279	1.037	GNN-CN	11.324	63.214
GCN	12.435	1.645	GCN-CN	12.523	64.352
SGNN	21.674	2.523	STGNN-CN	38.624	65.142
MLP-WOA	35.175	3.348	AHSTGNN-CN	45.427	69.341

**Fig. 24.** The average training time and average running time of each prediction model.

AHSTGNN-CN, STGNN and STGNN-CN, GNN and GNN-CN, and GCN-CN. After comparison, the RMSE of GNN, GCN, STGNN, and AHSTGNN after NWP correction decreased by 0.00226, 0.0015, 0.00159, and 0.00095 respectively compared to before correction, MAE decreased by 0.00214, 0.00225, 0.00091, and 0.00112 respectively, R² increased by 0.0142, 0.0111, 0.0154, and 0.0067 respectively, and accuracy increased by 0.23 %, 0.15 %, 0.15 %, and 0.09 % respectively, verifying the effectiveness of the method of NWP wind speed correction proposed in this paper.

3.2.2. Wind power prediction for wind farm cluster B

The time range of data for the large-scale WFC B is from March 1, 2020 to December 30, 2020. The information of NWP includes wind speed, wind direction, temperature, and humidity. The time resolution of WP and NWP is both 15 min. The information of NWP is input into the AHSTGNN model for training. And the proportions of the training set, validation set, and test set are 70 %, 10 %, and 20 %, respectively. The scale of input time of the AHSTGNN model is 24 h, and the scale of output time is also 24 h. At the same time, we use grid optimization combined with ten-fold cross validation to optimize the hyperparameters of each model in the training set, making the model have robust predictive performance and generalization ability. The important hyperparameters for the different prediction models are shown in Table 4 for wind farm cluster B.

To verify the effectiveness of the method proposed in this paper, it was compared with CNNLSTM, TCN, GRU, GNN, GCN, SGNN, MLP-WOA, LSTM-GOW, STGNN, AHSTGNN, GNN-CN, GCN-CN, and STGNN-CN. The predicted results and actual power values of 14 models are shown in the Fig. 19. After calculation, the RMSE, MAE, R², and accuracy rate of the 14 models are shown in Table 5 and Fig. 20. The Table 5 presents three types of comparisons such as ablation experiments, comparisons between various models, and comparisons of prediction accuracy before wind speed correction and after NWP wind speed correction.

(1) Ablation experiment

The algorithms involved in this section include AHSTGNN, STGNN, SGNN, and GCN. After comparison, GCN has the lowest prediction accuracy among the methods in this section. SGNN adds spatial attention mechanism compared to GCN, which can capture spatial correlation features between wind farms. Compared to GCN, RMSE and MAE are reduced by 0.00784 and 0.00779, respectively, and R² and AR are improved by 0.0279 and 0.79 %, respectively. Compared with SGNN, STGNN adds a time attention mechanism that can capture the temporal correlation of internal feature information of wind farms, further

improving the prediction accuracy of WP. Compared with SGNN, STGNN algorithm reduces RMSE and MAE by 0.0004 and 0.00098, and improves R^2 and AR by 0.0002 and 0.04 %, respectively. AHSTGNN has added adaptive dynamic adjustment function comparing with STGNN, which can adaptively match the changes in the topology of WFC. After comparing the prediction accuracy of AHSTGNN with STGNN, the RMSE and MAE decreased by 0.00081 and 0.0017 respectively, and the R^2 and AR increased by 0.0200 and 0.08 %, respectively. The above evaluation indicators further demonstrate the effectiveness of the model proposed in this paper.

(2) Comparison between various algorithms

The algorithms involved in this section include CNNLSTM, TCN, GRU, GNN, MLP-WOA, LSTM-WOA, and AHSTGNN. After comparison, the RMSE of the method proposed in this paper decreased by 0.01662, 0.01831, 0.01034, 0.01039, 0.0019, and 0.00269 compared to the other six methods, and the MAE decreased by 0.01535, 0.01728, 0.00982, 0.00945, 0.00165, and 0.00152 compared to the other six methods. The R^2 improved by 0.0606, 0.0592, 0.0593, 0.0562, 0.0289, and 0.0320 compared to the other six methods, and the accuracy improved by 1.67 %, 1.83 %, respectively. 1.04 %, 1.04 %, 0.19 %, and 0.27 %. Based on the above analysis, the model proposed in this article is superior to the compared algorithms, demonstrating the superiority of the method proposed in this article.

(3) Comparison of prediction accuracy after NWP correction

The algorithms involved in this section include AHSTGNN and AHSTGNN-CN, STGNN and STGNN-CN, GNN and GNN-CN, and GCN-CN. After comparison, the RMSE of GNN, GCN, STGNN, and AHSTGNN after NWP correction decreased by 0.00716, 0.00598, 0.00078, and 0.00067, MAE decreased by 0.00711, 0.00603, 0.00270, and 0.00164, R^2 increased by 0.0124, 0.0088, 0.0149, and 0.0042, and accuracy increased by 0.71 %, 0.60 %, 0.07 %, and 0.06 %, respectively, verifying the effectiveness of the method of NWP wind speed correction proposed in this paper.

3.3. Analysis and discussion

To further analyze the effectiveness of the method proposed in this article, the prediction results of two wind farm clusters are comprehensively analyzed. Table 6 shows the average changes in evaluation indicators for the prediction results of two wind farm clusters comparing AHSTGNN with GCN, SGNN, and STGNN during the ablation experiment.

According to Table 6 and Fig. 21, the prediction results of AHSTGNN have improved to varying degrees compared to those of GCN, SGNN, and STGNN. Meanwhile, the ranking of prediction accuracy from high to low for the four prediction methods is as follows: AHSTGNN, STGNN, SGNN, and GCN. GCN has limited ability to handle large-scale graphs and it is sensitive to changes in graph structure. GCN usually assumes that the topology of the graph is fixed. However, in practical applications, the topology of the graph is often dynamically changing. In contrast, SGNN introduces a spatial attention mechanism that can assign different weights to neighboring nodes of each node, allowing the model to focus more on neighbor information that is more closely related to the current node. Spatial attention mechanism can help to better capture the local structural features of the graph and enhance the ability of model to perceive local details. STGNN incorporates a time attention mechanism based on SGNN, which can identify key time points and dependencies between time steps in time series data, allowing the model to focus more on important time information to further improve prediction accuracy. The time attention mechanism assigns different weights to features at different time steps, enabling the model to more effectively extract and it fuses feature information in the time dimension. AHSTGNN further

introduces adaptive graph neural networks, which enables the model to dynamically adjust attention weights and aggregation strategies based on the characteristics of input data, thereby more accurately capturing complex relationships between spatiotemporal features. Adaptability enables the model to flexibly adjust its parameters and structure based on the distribution characteristics of data in order to achieve optimal performance, which enables it to automatically adapt to graph data of different scales and densities as well as the length and variation characteristics of different time series.

Table 7 and Fig. 22 show the average trend of the evaluation indicators obtained by comparing AHSTGNN with CNNLSTM, TCN, GRU, GNN, MLP-WOA, and LSTM-GOW of the results of WPP for two WFCs. It can be seen that the predictive performance of the method proposed in this article has been improved to varying degrees compared to the comparative algorithms. The method proposed in this article introduces temporal attention mechanism, spatial attention mechanism, and adaptive graph neural network. These mechanisms can effectively capture the local structural features of the graph, making the model more focused on neighbor information that is more closely related to the current node. In addition, the temporal attention mechanism can identify the dependency relationships between key time points and their time steps in time series data, thereby making the model more focused on important temporal information. Adaptability enables the model to dynamically adjust attention weights and aggregation strategies based on the characteristics of input data, thereby more accurately capturing the complex relationships between spatiotemporal features.

After applying the NWP correction method proposed in this article to GCN, GNN, STGNN, and AHSTGNN models, the performance of all four methods is improved to varying degrees compared to before NWP correction. Table 8 and Fig. 23 show the average variation trend of each prediction model for two wind farm clusters compared to their own evaluation indicators of predicted power after NWP correction. However, after comprehensive analysis, the AHSTGNN and STGNN models improved the accuracy of wind power prediction by less than 0.12 % after NWP wind speed correction, while the GNN and GCN models showed more significant improvements in prediction accuracy, increasing by 0.47 % and 0.38 %, respectively. The STGNN and AHSTGNN introduce spatial attention mechanisms, which can effectively capture the local structural features of the graph and improve the prediction accuracy of wind power by utilizing information from other relevant wind farms. Overall, the NWP correction method proposed in this article has limited accuracy improvement for prediction methods that can utilize relevant wind farm information, while it can significantly improve the accuracy of prediction methods that do not utilize other relevant feature information for wind farms.

3.4. Comparison of running time

To verify the practicality of the models proposed in this article, simulations and calculations were performed on the 14 models using a computer whose central processing unit is Inter Core i9-10900 K. After 50 calculations, the average training time and average running time of the 14 prediction models mentioned above were obtained. The average training time and average running time of the 14 prediction models are shown in the Table 9 and Fig. 24.

4. Conclusion

This paper proposes a short-term power prediction method based on NWP correction and the fusion of adaptive spatiotemporal graph feature information for WFC. The contributions of this paper are as follows:

- (1) This article proposes a new definition indicator – error correlation ratio, which is used to determine the similarity between two time series. The smaller the ECR, the higher the similarity between two time series segments.

- (2) This article proposes a wind speed correction method for NWP based on ECR to find two similar time periods. After comparison, the NWP correction method proposed in this paper reduced the average RMSE and MAE by 0.0029 and 0.0031, respectively, and increased the average R² and AR by 0.0148 and 0.28 %, respectively, verifying the effectiveness of the NWP wind speed correction method proposed in this paper.
- (3) This article proposes a short-term prediction model for WFC based on adaptive spatiotemporal graph feature information fusion. The adaptive spatiotemporal graph and dynamic graph are introduced into the WPP model to further explore the temporal and spatial features of WFC. After verification, compared with other models, the method proposed in this article reduced RMSE and MAE by an average of 0.0100 and 0.00956 respectively and improved R² and AR by 0.0440 and 1.01 % respectively in average.
- (4) In future work, the proposed model will be improved as follows: (1) The method of NWP wind speed correction mentioned in this article does not consider the data characteristics from other related wind farms in the wind farm cluster, resulting in limited correction performance of the method. (2) The model took a lot of time to train, so it is necessary to improve the performance of the prediction model. Meanwhile, it is recommended to use higher performance computers to train the model in order to reduce training time.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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